Mixing Self-Attention and Convolution: A Unified Framework for Multisource Remote Sensing Data Classification

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***Abstract*— Convolution and self-attention are two powerful techniques for multisource remote sensing (RS) data fusion that have been widely adopted in Earth observation tasks. However, convolutional neural networks (CNNs) are inadequate for fully mining contextual information and representing the sequence attributes of spectral signatures. In addition, the specific self-attention mechanism often comes with high-computational costs, which hinders its application in the field of RS. To overcome the above limitations, this article proposes a unified framework called "mixing self-attention and convolution network (MACN)" for comprehensive feature extraction and efficient feature fusion. First, the proposed MACN utilizes two adaptive CNN encoders (ACEs) to extract shallow convolutional features from multisource RS data. Second, taking the complexity and varying scales of RS data into account, the proposed mixing self-attention and convolution Transformer (MACT) layer achieves local and global multiscale perception through an elegant integration of self-attention and convolution. MACT can extract abundant spatial and high-dimensional information (e.g., spectral and elevation information) while maintaining minimal computational overhead compared with pure convolution or self-attention counterparts. Finally, a multisource cross-guided fusion (MCGF) module is designed to achieve deep fusion of multisource RS data features. MCGF utilizes a carefully designed cross-modal attention mechanism to capture the interaction between multisource data and aggregate contextual information. Extensive tests on six public RS datasets have shown that our method outperforms other multisource fusion models, delivering state-of-the-art (SOTA) results on multiple RS data fusion tasks without specific tuning. The source code of the proposed method will be available publicly at https://github.com/like413/MACN.**

***Index Terms*— Mix self-attention and convolution, multisource cross-guided fusion, multisource remote sensing (RS).**

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I. Introduction

With the rapid development and diversification of remote sensing (RS) sensor techniques, there have been ever-increasing amounts of multisource RS data acquired from different platforms, such as satellites, aircraft, and unmanned aerial vehicles [1], [2]. RS with multisensor devices has been systematically applied to monitoring land use and cover classification [3], disaster monitoring and response [4], climate and meteorological research [5], geological and resource exploration [6], urban planning [7], agriculture production [8], and other fields. Land-cover classification has long been an important but challenging task.

Most of the previous works rely on a single-sensor data, such as optical images, infrared images, hyperspectral images (HSIs), multispectral images (MSIs), light detection and ranging (LiDAR), and synthetic aperture radar (SAR). However, single-senor RS data have limitations in spatial resolution, spectral resolution, temporal resolution, and other factors that may not be sufficient to meet the requirements of all applications. To overcome these limitations, multisource RS data fusion technology has emerged. Multisource RS data fusion aims to integrate and fuse RS data from different sensors, bands, or times to obtain more comprehensive, accurate, and informative RS information. For example, HSI can effectively distinguish different materials due to their rich spectral features [9], SAR can be used for observation under adverse weather and lighting conditions by utilizing microwave signals [10], and LiDAR, which uses laser beams, can extract distance information of the ground and objects [11]. By selecting and effectively fusing data from different modes, we can obtain more comprehensive and multidimensional RS information, which, in turn, provides more accurate land-cover classification results.

Several classic machine learning algorithms have been proposed for land-cover classification, such as support vector machines (SVMs) [12], random forest (RF) [13], and K-nearest neighbors (KNN) [14]. However, these methods struggle to process data with high-dimensional information, resulting in limited classification performance. Principal component analysis (PCA) [15] is one of the most widely used methods for data dimensionality reduction, which successfully reduces the computation cost when processing high-dimensional RS data. However, it does not adequately consider spatial and structural information and still fails to

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achieve satisfactory accuracy. To address this issue, some researchers have attempted to extract spatial and structural features of RS data using digital image processing methods, including edge extraction [16], dilation and erosion [17], and image transformation [18]. To mine information from multisource RS data, Akhtar et al. [19] used nonparametric Bayesian sparse representation to sparsely code the image and fuse high-resolution images with low-resolution images. Zhang et al. [20] introduced sparse representation to learn over-complete dictionaries from the training images for image fusion, which is more stable and meaningful than typical multiscale transforms (MSTs) when representing source images. However, these methods heavily rely on prior knowledge of the RS data and handcrafted features, which may not be able to represent complex high-level semantic information, making it challenging to improve classification accuracy while maintaining the robustness of the algorithm.

Deep learning has emerged as a powerful tool for multisource RS data fusion. As one of the most used deep learning methods, convolutional neural networks (CNNs) can automatically learn hierarchical representations of RS data, allowing them to capture local spatial features. Depending on the fusion stage, CNN-based fusion methods can be categorized into pixel-level fusion, decision-level fusion, and feature-level fusion.

1) Pixel-level fusion methods directly process the original data. Hu et al. [21] proposed a multimodal data fusion algorithm that amplifies the influence of the existing training set by linking labeled data to unlabeled data through unsupervised techniques. Hong et al. [22] developed a deep encoder and decoder network to fuse multimodal information by forcing feature fusion and reconstructing multimodal inputs sequentially. To further improve the joint classification accuracy, Zhao et al. [23] used octave convolution to fuse the original data according to low-frequency and high-frequency information, retaining more discriminative information in RS data. Gaetano et al. [24] proposed a new deep learning architecture for simultaneously using panchromatic (PAN) and multispectral (MS) data sources to fuse with Markov random fields (MRFs), aiming to minimize information loss by utilizing the original form of PAN and MS information. The advantage of these methods lies in minimizing information loss. However, due to the typically large volume of original data, these pixel-level fusion methods unavoidably result in data redundancy and increased computational costs.
2) Decision-level fusion performs weighted judgment on the recognition results of different data types. To improve the quality of the predicted accuracy, Paisitkriangkrai et al. [25] fused classification map obtained by CNN with that obtained by RF classifier trained by handcrafted features. Audebert et al. [26] fused two images obtained from a pretrained residual fusion network. Sherrah [27] trained several CNNs that are used as activation mappings for each layer of the main model and as the input for fusion. To solve the problem caused by the imbalance among different features, Li et al. [28] developed the selective feature assignment module (SFAM) to adaptively assign weights to different discriminative features. However, decision-level fusion usually requires processing and integrating data from multiple sensors, which can result in high-computational complexity and resource requirements. It also lacks flexibility and may not fully leverage the complementary advantages of multisource RS data.
3) Feature-level fusion involves fusing the features of multisource data. The simplest method is to stack d-dimensional features by concatenation, such as pseudo-siamese convolutional neural network (PSCNN) [29] and multisource remote sensing data classification (MRSDC) [30], which may not take full advantage of the properties of each feature and can exhibit unstable performance. Khodadadzadeh et al. [31] pointed out that simple feature concatenation or stacking can increase the input features of very high dimension, exacerbating the Hughes effect. Fusing multisource RS data by coupling to reduce redundant details and preserve distinctive geometric features remains challenging. Wu et al. [32] proposed a cross-channel reconstruction module to learn more compact fusion representations of different RS data, achieving more efficient information exchange. Hong et al. [33] proposed a general multimodal deep learning framework for RS data classification, which extensively studies and summarizes various multimodal fusion strategies and training methods, providing comprehensive guidance for multimodal feature learning and cross-modal data processing.

Although the existing CNN-based methods have significantly improved the classification performance of multisource RS data fusion, they still have some inherent drawbacks. For example, due to the restriction of convolutional kernels, CNNs cannot capture rich contextual information; CNNs are good at spatial feature representation, but it is difficult for them to process high-dimensional information. Considering these reasons, researchers have explored other deep learning architectures. Wu and Prasad [34] used recurrent neural network (RNN) to extract contextual spectral features from the previous feature. Liu et al. [35] pointed out that graph CNNs (GCNNs) can utilize the correlations between adjacent land covers and effectively reduce computational complexity. Luo et al. [36] combined residual and dense connection ideas to propose fusion generative adversarial network (FusGAN), which further preserves the spectral–spatial information in the fusion image.

More recently, Transformer has gradually attracted the attention of researchers for its ability to globally model contextual information at different positions in a sequence. It is initially used in natural language processing [37] and has been applied to computer vision [38], where some works successfully apply it to RS classification tasks. Hong et al. [39] proposed a novel backbone network called SpectralFormer to extract sequence information of spectral features. Qing et al. [40] used spectral attention mechanism and self-attention mechanism to enhance the classification performance of RS data. However, although these works successfully capture spectral features

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using Transformers, they ignore spatial information in RS data. Tang et al. [41] constructed a multiscale Transformer to capture rich contextual information from multilevel features. Zhao et al. [42] designed a novel dual-branch network to capture and learn spectral–spatial features from HSI data and elevation features from LiDAR data.

Based on the aforementioned discussion, our objective is to enhance the model's capability to perceive multiple scales and effectively extract high-dimensional information from RS data. In addition, we aim to minimize the model's parameter count and reduce computational costs as much as possible. To address these requirements, we propose a unified framework named mixing self-attention and convolution network (MACN) in this article. MACN is devised for comprehensive multisource RS data feature extraction and effective feature fusion. First, we utilize two adaptive CNN encoders (ACEs) to extract shallow convolutional features from multisource RS data. Second, the introduction of the mixing self-attention and convolution Transformer (MACT) layer aims to comprehensively capture the local–global multiscale information hidden within the RS data. By analyzing the convolutional and self-attention structures and reengineering them, the MACT layer not only enables the exploration of rich spatial and high-dimensional information (e.g., spectral and elevation information) in multisource RS data but also reduces computational costs. Accordingly, it is more suitable for RS scene classification. Next, a simple yet effective multisource cross-guided fusion (MCGF) module is designed to cross-guide and aggregate contextual information from two branches of RS data features. Therefore, MACN maximizes the utilization of complementary advantages in multisource RS data, enabling efficient fusion and enhancing the overall performance. Finally, the two fused features from the MCGF module are added together and fed into a simple linear classifier to obtain the labels for the pixels.

The major contributions of this article are summarized as follows.

1) We propose an ACE that can select a suitable convolutional feature extraction network based on the shape of the input data. Moreover, we have redesigned the image encoder specifically for images with spectral information to better preserve spatial structure and spectral information.
2) We propose the MACT layer, which can effectively capture local–global multiscale information from RS data. By analyzing the convolutional and self-attention structures and reengineering them, the MACT layer reduces computational costs and is more robust and suitable for RS scene classification.
3) A simple yet effective MCGF module is proposed to cross-guide and aggregates the features from two branches of RS data. Therefore, MACN maximizes the utilization of complementary advantages in multisource RS data, enabling efficient fusion and enhancing the overall performance.
4) We quantitatively evaluate the classification performance of the proposed MACN on six representative multisource RS datasets. The encouraging results demonstrate that the proposed MACN is effective for multisource RS scene classification.

The rest of this article is organized as follows. Section II elaborates on the proposed method. The extensive experiments and analyses are presented in Section III. Finally, the conclusions of this work and some discussions on possible future research directions are drawn in Section IV.

II. Methodology

The MACN framework aims at pixel-level classification by fusing multisource RS data. The framework of the proposed MACN is illustrated in Fig. 1. It consists of five stages: 1) multisource RS data preprocessing in Section II-A; 2) ACE extracts shallow convolution features in Section II-B; 3) MACT layer realizes local–global multiscale modeling in Section II-C; 4) MCGF for cross fusion in Section II-D; and 5) classifier to obtain the classification map using the comprehensive features in Section II-E.

It is worth mentioning that due to the presence of the ACE, MACN can perform classification tasks on any paired multisource RS data, such as optical image, infrared image, SAR, LiDAR, HSI, MSI, and so on. For convenience, we will use HSI data and LiDAR data throughout the rest of this article to describe our framework.

A. Multisource RS Data Preprocessing

For HSI/LiDAR fusion, the HSI data $X_H \in \mathbb{R}^{h\times w\times c}$ and the LiDAR data $X_L \in \mathbb{R}^{h\times w}$ are acquired from different sensors, covering the same area of the Earth's surface, where h and w represent the two spatial sizes and c is the number of HSI spectral bands. The HSI data typically consists of a large number of spectral bands, which provide valuable information. However, this can increase the complexity of computation and storage requirements. To reduce the spectral dimension, PCA is used to extract the first d principal components of X_H, maintaining the spatial dimensions but lowering the number of bands from c to d. After PCA, the HSI data X_H are transformed into $X_H^{\text{pca}} \in \mathbb{R}^{h\times w\times d}$.

Next, the dimensionally reduced data are divided into patches of size $p \times p$ to obtain $X_H^P \in \mathbb{R}^{p\times p\times d}$ and $X_L^P \in \mathbb{R}^{p\times p}$. For edge pixels, padding of size less than p to the same size. Based on the ground-truth map, after removing the pixel blocks with a label of zero, the remaining samples are divided into a training set and a test set.

B. Adaptive CNN Encoder

Because of its ability to extract edges, textures, and other local features from images, CNNs are effective for 2-D RS data. In addition, 3-D CNNs have been proven to locally model high-dimensional information in 3-D RS data. However, the use of 3-D CNNs increases the computation cost of processing RS data. Therefore, in the proposed framework, two ACEs are used to extract spectral–spatial information from HSI data and elevation information from LiDAR data, respectively.

As shown in Fig. 2, the ACE consists of two parts: the gate and the dual-branch CNN encoder. The gate adaptively determines the usage of the CNN encoder based on the input

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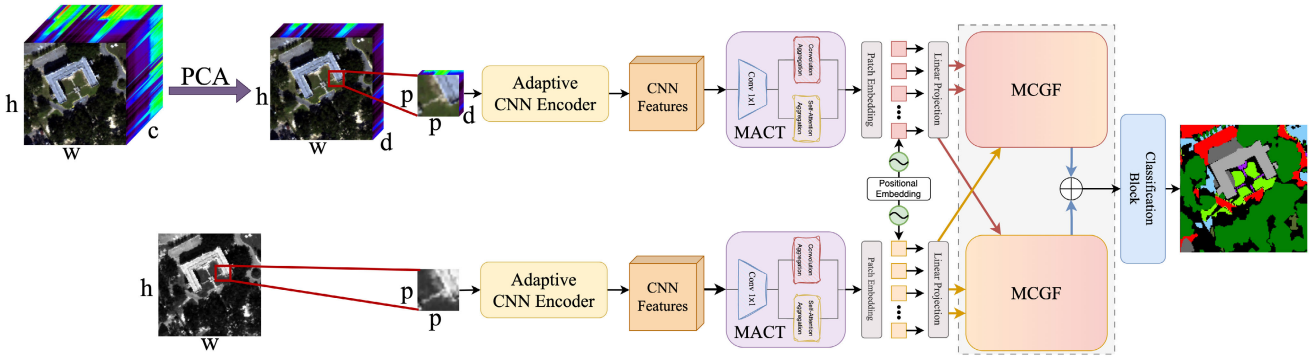


Fig. 1. Proposed MACN classification framework.

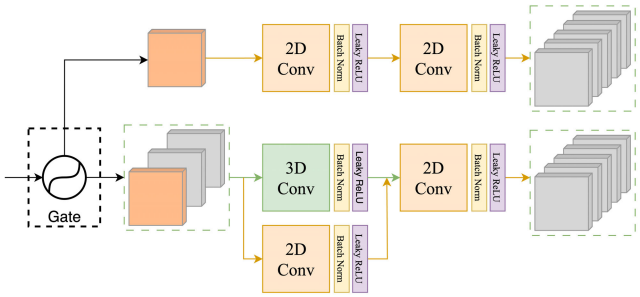


Fig. 2. Graphical illustration of the ACE.

data dimension. The upper branch uses two Conv-2D layers to extract LiDAR elevation features X_L^{conv}

$$\text{Conv2D}(1, 16) \Rightarrow \text{Conv2D}(16, 64) \tag{1}$$

where $\text{Conv2D}(c_1, c_2)$ represents a 2-D convolutional layer, and c_1 and c_2 represent the number of channels of input and output data, respectively.

Unlike the LiDAR processing, the lower branch uses two different convolutional layers to extract the HSI features X_H^{conv}: first layer uses Conv-2D as the residual edge of Conv-3D, and the second layer uses a regular Conv-2D

$$\text{Reshape}(\text{Conv3D}(d, s)) + \text{Conv2D}(1, 64) \Rightarrow \text{Conv2D}(64, 64) \tag{2}$$

where $\text{Conv3D}(\cdot)$ represents a 3-D convolutional layer, d and s represent the number of HSI spectral bands of input and output data, and $\text{Reshape}(\cdot)$ represents the transformation of 3-D data into 2-D data. To regularize and speed up the training process, a batch normalization layer and a leaky rectified linear unit (LeakyReLU) function are consecutively applied after each convolutional layer.

C. Mixing Self-Attention and Convolution Transformer

The MACT layer is used to perform local–global multiscale modeling on the HSI/LiDAR shallow CNN features. The MACT layer not a simple concatenation or parallelization of CNNs and Transformers. Instead, it is constructed by decomposing and reassembling the execution modes of convolution and self-attention. The MACT layer enables the exploration of abundant spatial and high-dimensional information (e.g., spectral and elevation information) in multisource RS data and also reduces computational costs.

1) Revisiting Convolution: Given tensors $X \in \mathbb{R}^{H\times W\times C_{\text{in}}}$ and $Y \in \mathbb{R}^{H\times W\times C_{\text{out}}}$ as the input and output feature maps, where H and W represent the height and width, respectively, and C_{in} and C_{out} are the input and output channel size, respectively. For simplicity, we apply a standard convolution with the stride $s = 1$ and kernel $K \in \mathbb{R}^{C_{\text{out}}\times C_{\text{in}}\times k\times k}$ to the X, where k is the kernel size. Consider $X_{ij} \in \mathbb{R}^{C_{\text{in}}}$ and $Y_{ij} \in \mathbb{R}^{C_{\text{out}}}$ as the feature tensors of pixel (i, j) corresponding to X and Y. Then, the standard convolution can be formulated as follows:

$$Y_{ij} = \sum_{m,n} X_{ij}^{m,n} = \sum_{m,n} K_{m,n} X_{i+m-\lfloor k/2\rfloor, j+n-\lfloor k/2\rfloor} \tag{3}$$

where $K_{m,n} \in \mathbb{R}^{C_{\text{out}}\times C_{\text{in}}}$, $m, n \in \{0, 1, \ldots, k-1\}$, represents the kernel weights with the kernel position (m, n).

For convenience, we define the shift operation as follows:

$$\begin{aligned} \forall i, j, \tilde{F} &= \text{Shift}(F, \triangle x, \triangle y) \\ &= F_{i+\triangle x, j+\triangle y} \end{aligned} \tag{4}$$

where $\triangle x$ and $\triangle y$ correspond to the horizontal and vertical displacements, respectively. Then, $X_{ij}^{m,n}$ can be rewritten as follows:

$$X_{ij}^{m,n} = \text{Shift}(K_{m,n} X_{i,j}, m - \lfloor k/2\rfloor, n - \lfloor k/2\rfloor). \tag{5}$$

As a result, the standard convolution can be summarized as two stages as follows:

$$\textbf{Stage I}: \tilde{Y}_{ij}^{m,n} = K_{m,n} X_{ij} \tag{6}$$

$$\textbf{Stage II}: Y_{ij}^{m,n} = \text{Shift}(\tilde{Y}_{i,j}, m - \lfloor k/2\rfloor, n - \lfloor k/2\rfloor) \tag{7}$$

$$Y_{ij} = \sum_{m,n} Y_{ij}^{m,n}. \tag{8}$$

2) Revisiting Self-Attention: Similar to the previous operations, $X_{ij} \in \mathbb{R}^{C_{\text{in}}}$ and $Y_{ij} \in \mathbb{R}^{C_{\text{out}}}$ represent the corresponding tensor of pixel (i, j). Then, the standard attention can be formulated as follows:

$$Y_{ij} = \bigcup_{l=1}^{N} \left\{ \sum_{m,n\in N_k(i,j)} A\left(q_{ij}^{(l)}, k_{mn}^{(l)}, v_{mn}^{(l)}\right) \right\} \tag{9}$$

where $\bigcup$ is the concatenation of the outputs of N attention heads, $N_k(i, j)$ represents a local region of pixels with spatial extent k centered around (i, j), and $A(\cdot)$ is the corresponding attention weight with regard to the features within $N_k(i, j)$.

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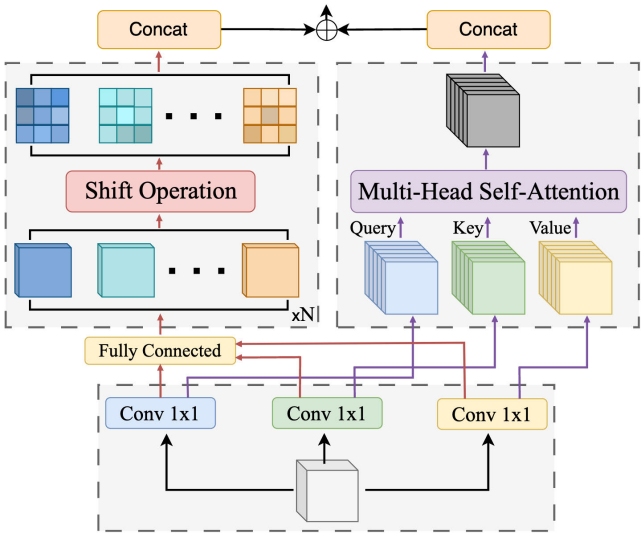


Fig. 3. Graphical illustration of the MACT.

The $A(\cdot)$ in (9) is computed as follows:

$$\text{Attention}(\cdot) = \text{Softmax}\left(\frac{\left(q_{ij}^{(l)}\right)^T k_{mn}^{(l)}}{\sqrt{d}}\right) v_{mn}^{(l)} \tag{10}$$

where d is the dimension of $q_{ij}^{(l)}$.

Also, multihead self-attention can be decomposed into two stages and reformulated as follows:

$$\textbf{Stage I}: q_{ij}^{(l)} = W_q^{(l)} X_{ij}, \quad k_{ij}^{(l)} = W_k^{(l)} X_{ij}, \; v_{ij}^{(l)} = W_v^{(l)} X_{ij} \tag{11}$$

$$\textbf{Stage II}: Y_{ij} = \bigcup_{l=1}^{N} \left\{ \sum_{m,n \in N_k(i,j)} A\left(q_{ij}^{(l)}, k_{mn}^{(l)}, v_{mn}^{(l)}\right) \right\} \tag{12}$$

where W_q, W_k, and W_v are the projection matrices for queries, keys, and values, respectively.

3) Rediscussing Self-Attention and Convolution: After decomposing the self-attention and convolution modules as described above, we observe that these two stages share many similarities. Stage I is a feature learning module, where self-attention and convolution modules share the same operations by performing 1×1 convolutions to project features into deeper spaces. Therefore, these operations can be shared to reduce computational overhead. Stage II corresponds to the procedure of feature aggregation, and both modules are lightweight and do not acquire additional learnable parameters.

4) Integrating Self-Attention and Convolution: As shown in Fig. 3, the proposed MACT also includes two stages. At Stage I, input feature is projected by three 1×1 convolutions and reshaped into C pieces, respectively. Thus, we obtain a rich set of intermediate features containing $3 \times C$ feature maps. At Stage II, it consists of two different branches of operations on self-attention and convolution. For the self-attention branch, we gather the intermediate features into N groups, where each group contains three pieces of features, one from each 1×1 convolution. The corresponding three feature maps serve as queries, keys, and values. For the convolutional branch with kernel size k_c, we reorganize the number of channels C of the above three feature maps into N groups, respectively. Subsequently, we concatenate N groups and use linear projection to expand $3 \times N$ channels to k_c^2 channels. Then, a reorganization of the convolution method is achieved by shifting and aggregating the generated features. Finally, outputs from both branches are added together

$$F_{\text{MACT}} = \alpha F_{\text{attention}} + \beta F_{\text{convolution}} \tag{13}$$

where α and β are two balance parameters, which are learned during the training process. We use MACT to obtain F_{MACT}^{H} and F_{MACT}^{L} for HSI features X_H^{conv} and LiDAR features X_L^{conv}, respectively.

TABLE I

FLOPS AND PARAMETERS FOR DIFFERENT MODULES AT TWO STAGES. NUMBERS IN RED CORRESPOND TO THE ADDITIONAL PARAMS/ FLOPS INTRODUCED BY MACT

Module	Stages	#Param.	FLOPs
Convolution	I	$k_c^2C^2$	$k_c^2C^2$
	II	0	k_c^2C
Self-Attention	I	$3C^2$	$3C^2$
	II	0	$2k_a^2C$
MACT(Ours)	I	$3C^2$	$3C^2$
	II	$3k_c^2N$	$3k_c^2C+k_c^2C+2k_a^2C$

5) Computational Cost of MACT: For better comparison, we analyze the floating-point operations (FLOPs) and the number of parameters at each stage and summarize in Table I. C represents input and output channel, k_c and k_a represent kernel size for convolution and self-attention, respectively, and N represents the number of attention heads. It is shown that theoretical FLOPs and parameters at Stage I of convolution have quadratic complexity with regard to the channel size C, while the computational cost for Stage II is linear to C, and no additional training parameters are required. A similar trend is also found for the self-attention module, where all training parameters are preserved at Stage I. The computational cost and training parameters of MACT at Stage I are the same as self-attention and lighter than traditional convolution (e.g., 3×3 conv). At Stage II, MACT introduces additional computation overhead with a light linear projection layer, whose computation complexity is linear with regard to channel size C and comparably minor with Stage I.

D. Multisource Cross-Guided Fusion

In order to fully utilize the complementary information from multisource RS data and achieve deep fusion of features from different modes, an MCGF is carefully designed to fuse HSI and LiDAR data features. Specifically, we use the features from one data mode as the primary features and fuse them with features from the other mode to obtain the preliminary fused features. These preliminary fused features are then used to guide the primary features, which are fused with the original primary features again to obtain the secondary fusion result. Due to the strategy of guiding before fusing, the semantic discrepancy between the two modalities of data is reduced, resulting in a more robust fusion. In our framework, the

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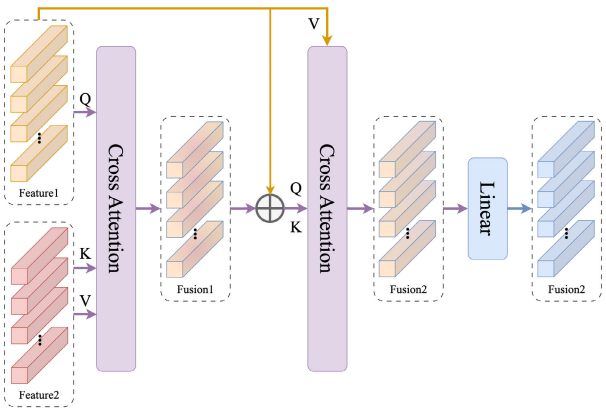


Fig. 4. Graphical illustration of the MCGF.

features of HSI and LiDAR data are sequentially used as primary features for cross-guided fusion to obtain two fused features. Finally, the two features are added together to obtain the final fused result.

Fig. 4 shows the graphical illustration of the MCGF layer. We select F_1 as the primary feature and linear project to obtain the query Q_1 and linear projection of feature F_2 from another modality to get key K_1 and value V_1, as shown in the following formula:

$$Q_1 = W_{q_1} F_1, \quad K_1 = W_{k_1} F_2, \quad V_1 = W_{v_1} F_2 \tag{14}$$

where W_{q_1}, W_{k_1}, and W_{v_1} are learnable weights, and d_{K_1} represents the dimension of K_1.

Next, we use the cross-modal attention mechanism to compute Q_1, K_1, and V_1 to obtain the preliminary fused feature T_1, and the cross-modal attention mechanism can be expressed as follows:

$$\text{CrossAttention}(\cdot) = \text{Softmax}\left(\frac{Q_i^T K_i}{\sqrt{d_{K_i}}}\right) V_i. \tag{15}$$

T_1 can be obtained by the following equation:

$$T_1 = \text{CrossAttention}(Q_1, K_1, V_1). \tag{16}$$

Then, we use the preliminary fused feature T_1 to guide the primary feature F_1 to reduce the semantic discrepancy between multisource RS data. Specifically, we linear mapping F_1 to obtain the value V_2 and then add F_1 and T_1 to linear mapping to obtain the query Q_2 and key K_2, as shown in the following formula:

$$Q_2 = W_{q_2}(T_1 + F_1), \quad K_2 = W_{k_2}(T_1 + F_1), \quad V_2 = W_{v_2} F_1 \tag{17}$$

where W_{q_2}, W_{k_2}, and W_{v_2} are learnable weights.

Finally, we feed Q_2, K_2, and V_2 into the cross-modal attention mechanism for fusion to obtain secondary fusion feature T_2, which can be obtained by the following equation:

$$T_2 = \text{CrossAttention}(Q_2, K_2, V_2). \tag{18}$$

To facilitate subsequent classification, we adjust the dimension of T_2 through linear projection to obtain T_3, as follows:

$$T_3 = \text{Linear}(T_2) \tag{19}$$

where T_3 is the fused feature after cross-guided fusion.

Algorithm 1 MACN

Require: HSI data $X_H \in R^{h \times w \times c}$, LiDAR data $X_L \in R^{h \times w}$, ground-truth data $Y \in R^{h \times w}$.

Ensure: Classification map **R**.

1: Initialize all weights and bias terms.
2: Obtain the X_H^{pca} after PCA transform.
3: Create all sample patches X_H^P and X_L^P from X_H^{pca} and X_L, and divide them into a training set and a test set.
4: **for** *epoch* $<$ *epochs* **do**
5: Perform two ACEs to extract CNN features X_H^{conv} and X_L^{conv} from X_H^P and X_L^P by Eqs. (1) and (2).
6: Perform two MACT layers to extract local-global multi-scale features F_{MACT}^H and F_{MACT}^L from X_H^{conv} and X_L^{conv} by Eq. (13).
7: Perform the MCGF layer to obtain T_3^{HL} and T_3^{LH} by Eqs. (20) and (21).
8: Combined T_3^{HL} and T_3^{LH} as Eq. (22) with two weighting factors and obtain the final fusion result F_{end}.
9: Input the F_{end} to the Classifier of MACN framework as Eq. (22).
10: Use a softmax function to identify the labels.
11: **end for**
12: Obtain classification map **R**.

We obtain T_3^{HL} and T_3^{LH} for HSI and LiDAR data features through MCGF, respectively, as shown in the following formula:

$$\left.\begin{array}{r} F_{\text{HSI}} \text{ as } F_1 \\ F_{\text{LiDAR}} \text{ as } F_2 \end{array}\right\} \Rightarrow T_3^{HL} \tag{20}$$

$$\left.\begin{array}{r} F_{\text{LiDAR}} \text{ as } F_1 \\ F_{\text{HSI}} \text{ as } F_2 \end{array}\right\} \Rightarrow T_3^{LH}. \tag{21}$$

To date, we have the LiDAR-guided HSI feature T_3^{HL} and the HSI-guided LiDAR feature T_3^{LH}. To improve classification accuracy, we perform decision-level fusion by adding the fused features from the multisource RS data, as follows:

$$F_{\text{end}} = \lambda T_3^{HL} + \mu T_3^{LH} \tag{22}$$

where F_{end} is our final fusion result, and λ and μ are two balance parameters, which are learned during the training process.

E. Classifier of MACN Framework

The F_{end} is fed into a multilayer perceptron (MLP) layer for the final classification. The MLP includes two linear layers with a Gaussian error linear unit (GELU) operation and a softmax classify layer to predict the probability distribution. However, there is an issue of imbalanced positive and negative samples in RS data. To address this issue, we introduce a dynamic scaling factor that can dynamically reduce the weights of easily distinguishable samples during the training process, thereby quickly focusing the center of gravity on those difficult-to-distinguish samples. Our loss function is as follows:

$$\mathcal{L}_{\text{class}} = -\alpha(1 - p_t)^{\gamma} \log(p_t) \tag{23}$$

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where p_t is the predicted probability of softmax output, and α and γ are the balanced factors.

The overall training process of the proposed MACN is shown in Algorithm 1.

III. Experiments and Analysis

To validate the effectiveness of the proposed network, experiments are conducted on six datasets. Section III-A introduces the datasets used for experiments that are presented. The details of the experimental setup are described in Section III-B, and the corresponding analyses are performed in Section III-C. In Section III-D, the experimental results of MACN are presented, along with a comparison with other methods. Ablation experiments are performed in Section III-E.

A. Data Description

Considering the diversity of datasets, we choose six RS datasets that vary in the volume of data: Missouri University and University of Florida (MUUFL) Gulfport [43], Houston2013 [31], Houston2018 [44], Trento [45], Augsburg [46], and Berlin [47].

1) MUUFL Dataset: Collected in November 2010 using a hyperspectral imager sensor through a reflective optics system at the Gulf Park campus of The University of Southern Mississippi, Long Beach, MS, USA. The dataset consists of a group of HSI and LiDAR data for the same area, with a pixel size of 325 × 220 and a spatial resolution of 2.5 m. The HSI data contain 64 spectral bands with a spectral resolution range from 375 to 1050 nm. The spatial resolution of the HSI is 0.54 × 1.0 m, while the digital surface model (DSM) based on LiDAR has a resolution of 0.60 × 0.78 m. This dataset depicts 11 urban land-cover classes. Fig. 6(a)–(c) visualizes a pseudo-color composite image of the HSI data, a gray-scale image of the LiDAR data, and a ground-truth map.

2) Houston2013 Dataset: Collected by the National Center for Airborne Laser Mapping (NCALM) with funding from the National Science Foundation (NSF) in 2012. The dataset covers the University of Houston campus and surrounding urban areas in Houston, TX, USA, and serves as the dataset for the 2013 IEEE geoscience and remote sensing society (GRSS) Data Fusion Contest. The dataset consists of a group of HSI and LiDAR data for the same area, with a pixel size of 349 × 1905 and a spatial resolution of 2.5 m. The HSI data contain 144 spectral bands with a spectral resolution range from 380 to 1050 nm. The LiDAR data consist of elevation in meters above sea level. This dataset depicts 15 urban land-cover classes. Fig. 7(a)–(c) visualizes a pseudo-color composite image of the HSI data, a gray-scale image of the LiDAR data, and a ground-truth map.

3) Houston2018 Dataset: Collected by the Hyperspectral Image Analysis Laboratory and the National Center for Airborne Laser Mapping (NCALM) in 2017. The dataset covers the University of Houston campus and surrounding urban areas in Houston, TX, USA, and serves as the dataset for the 2018 IEEE GRSS Data Fusion Contest. The dataset consists of MS-LiDAR, HSI, and MS-optical RS data, each containing 7, 48, and 3 channels, respectively. The HSI data cover a 380–1050-nm spectral resolution range. The laser wavelengths of three LiDAR sensors are 1550, 1064, and 532 nm. This dataset depicts 20 urban land-cover classes. Fig. 8(a)–(c) visualizes a pseudo-color composite image of the HSI data, a pseudo-color image for MS-LiDAR data, and a ground-truth map.

4) Trento Dataset: Captured in a rural area south of Trento, Italy. The dataset consists of a group of HSI and LiDAR data for the same area, with a pixel size of 600 × 166 and a spatial resolution of 1 m. The HSI data contain 63 spectral bands with a spectral resolution range from 420 to 989 nm. The LiDAR data consist of elevation in meters above sea level. This dataset depicts six rural land-cover classes. Fig. 9(a)–(c) visualizes a pseudo-color composite image of the HSI data, a gray-scale image of the LiDAR data, and a ground-truth map.

5) Augsburg Dataset: Captured over the city of Augsburg, Germany. The dataset consists of a group of HSI and dual-Pol PolSAR data for the same area, with a pixel size of 332 × 485 and a spatial resolution of 30 m. The HSI data contain 140 spectral bands with a spectral resolution range from 400 to 2500 nm. The dual-Pol [vertical transmit, vertical receive (VV)–vertical transmit, horizontal receive (VH)] SAR image contains four features, which are VV intensity, VH intensity, and real and imaginary parts of the off-diagonal elements of the PolSAR covariance matrix. This dataset depicts seven different land-cover classes. Fig. 10(a)–(c) visualizes a pseudo-color composite image of the HSI data, a pseudo-color composite image of the PolSAR data, and a ground-truth map.

6) Berlin Dataset: Captured over the urban and rural Berlin, Germany. The dataset consists of a group of SAR and HSI data for the same area, with a pixel size of 797 × 220 and a spatial resolution of 30 m. The HSI data contain 244 spectral bands with a spectral resolution range from 400 to 2500 nm. This dataset depicts eight different land-cover classes. Fig. 11(a)–(c) visualizes a pseudo-color composite image of the HSI data, a image of the SAR data, and a ground-truth map.

The names of the land-cover categories, the number of training samples, and the number of test samples for these six datasets are listed in Table II.

B. Experimental Settings

1) Evaluation Indicators: Three widely used evaluation metrics, namely, overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa), are used to quantitatively assess the classification performance of the proposed framework and other existing models.

2) Configurations: The experiments with the proposed method and other deep-learning methods were all implemented in the PyTorch platform, using a DELL EMC DSS8440 server with an Inter Xeon Silver 4210R 2.4-GHz CPU, 128-GB RAM, and an NVIDIA GeForce RTX 2080Ti GPU with 11 GB of VRAM. To optimize the network, the Adam optimizer was selected as the initial optimizer. For the training stage, the mini-batch size and the number of training epochs were set to 64 and 100, respectively.

C. Parameter Analysis

Before conducting the comparative experiment, we analyzed several hyperparameters that may affect the classification

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TABLE II

TRAINING AND TEST SAMPLE NUMBERS FOR MUUFL, HOUSTON2013, HOUSTON2018, TRENTO, AUGSBURG, AND BERLIN

No.	MUUFL dataset			Houston2013 dataset			Houston2018 dataset			Trento dataset			Augsburg dataset			Berlin dataset		
	Class Name	Training	Test	Class Name	Training	Test	Class Name	Training	Test	Class Name	Training	Test	Class Name	Training	Test	Class Name	Training	Test
C01	Trees	150	23096	Healthy Grass	198	1053	Healthy Grass	39196	20000	Apple Trees	129	3905	Forest	675	12832	Forest	443	54511
C02	Mostly Grass	150	4120	Stressed Grass	190	1064	Stressed Grass	130008	20000	Buildings	125	2778	Residential Area	1516	28813	Residential Area	423	268219
C03	Mixed Ground Surface	150	6732	Synthetic Grass	192	505	Artificial Turf	2736	20000	Ground	105	374	Industrial Area	192	2659	Industrial Area	499	19067
C04	Dirt and Sand	150	1676	Trees	188	1056	Evergreen Trees	54322	20000	Woods	154	8969	Low Plants	1342	25515	Low Plants	376	58906
C05	Road	150	6537	Soil	186	1056	Deciduous Trees	20172	20000	Vineyard	184	10317	Allotment	28	547	Soil	331	17095
C06	Water	150	316	Water	182	143	Bare Earth	18064	20000	Roads	122	3052	Commercial Area	82	1563	Allotment	280	13025
C07	Buildings Shadow	150	2083	Residential	196	1072	Water	1064	1628				Water	76	1454	Commercial Area	298	24526
C08	Buildings	150	6090	Commercial	191	1053	Residential	158995	20000							Water	170	6502
C09	Sidewalk	150	1235	Road	193	1059	Non-Residential	894769	20000									
C10	Yellow Curb	150	33	Highway	191	1036	Roads	183283	20000									
C11	Cloth Panels	150	119	Railway	181	1054	Sidewalks	136035	20000									
C12				Paking Lot1	192	1041	Crosswalks	6059	5345									
C13				Paking Lot2	184	285	Major Thoroughfares	185438	20000									
C14				Tennis Court	181	247	Highways	39438	20000									
C15				Running Track	187	473	Railways	27748	11232									
C16							Paved Parking Lots	45932	20000									
C17							Unpaved Parking Lots	587	3524									
C18							Cars	26289	20000									
C19							Trains	21479	20000									
C20							Stadium Seats	27296	20000									
-	Total	1650	52037	Total	2832	12197	Total	1859825	321729	Total	819	29395	Total	3911	74383	Total	2820	461851

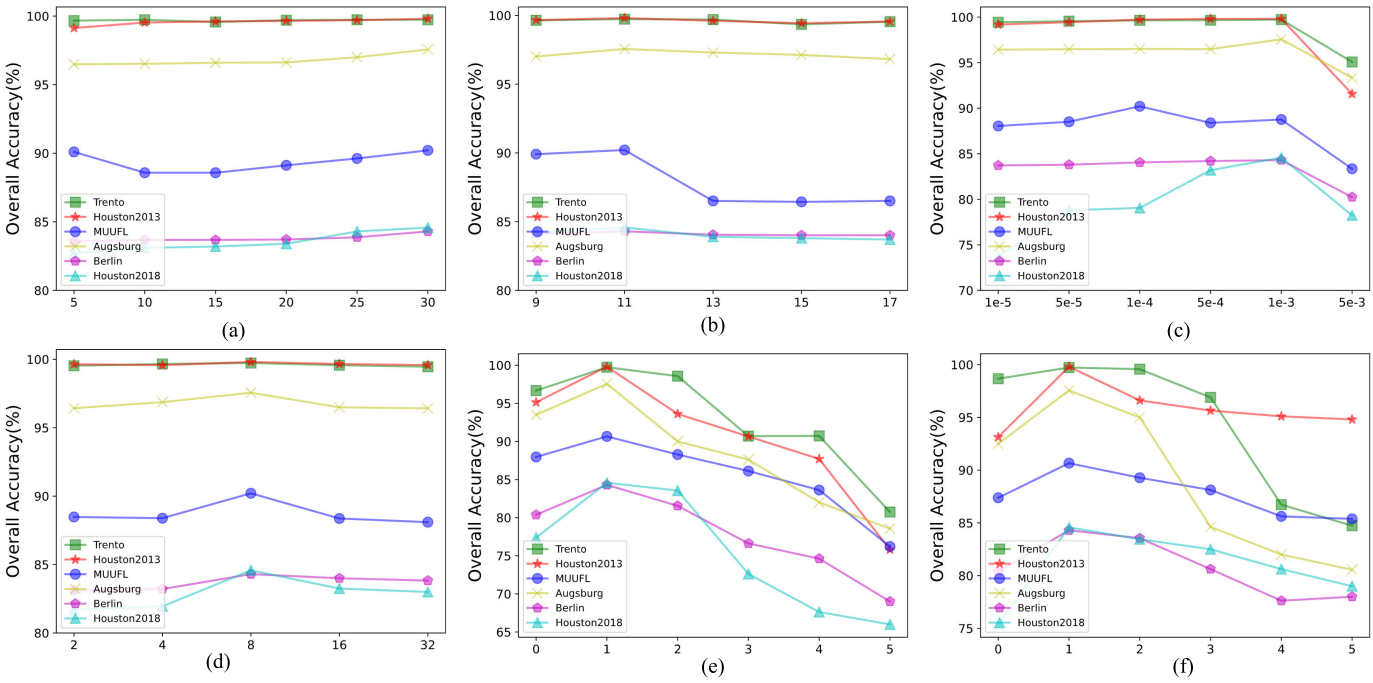


Fig. 5. Influence of different parameters on the OA. (a) Spectral dimension. (b) Patch size. (c) Learning rate. (d) Numbers of attention heads. (e) MACT layers. (f) MCGF layers.

performance, including the dimensionality of spectral data d, the patch size p of input data patches, the learning rate lr, the number of multiple attention heads head_num, and the attention layers used in MACN.

1) Spectral Dimension: We selected different spectral dimension d from the set $\{5, 10, 15, 20, 25, 30\}$ for comparison. Fig. 5(a) shows the OA performance of MACN on six datasets with different values of d, and we found that the classification performance is the best when $d = 30$.

2) Patch Size: Similar to the analysis of spectral dimension, we selected patch size p from the set $\{9, 11, 13, 15, 17\}$ to evaluate its effect. Fig. 5(b) shows the influence of different values of p on OA. We found that the best OA value was achieved on all six datasets when $p = 11$.

3) Learning Rate: Learning rate is an important hyperparameter in deep learning models, as it determines how quickly the objective function can reach its minimum. In our experiment, we selected the learning rate lr from the set {1e-5, 5e-5, 1e-4, 5e-4, 1e-3, 5e-3} to evaluate its effect. Fig. 5(c) shows the OA of MACN on six datasets with different learning rates. We found that the optimal learning rate for the MUUFL dataset is $1e-4$, while for the Houston2013, Houston2018, Trento, Augsburg, and Berlin datasets are $1e-3$.

4) Attention Heads: Both MACT and MCGF use multiple attention heads to learn the correspondence between different representation subspaces, where each head corresponds to an independent subspace with respect to different feature representations. Therefore, the number of attention heads used can affect the final feature representation capability of the Transformer and, hence, affect the classification performance. We selected different values of head_num from the set $\{2, 4, 8, 16, 32\}$ to evaluate their impact on OA. As shown in 5(d), the optimal OA value was achieved on all six datasets when head_num $= 8$.

5) Attention Layers: MACT and MCGF are the main components of the proposed model, and the number of layers in MACT and MCGF significantly affects the model's performance. In our experiment, we selected different numbers of

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TABLE III
CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE MUUFL DATASET

No.	Class Name	SVM [12]	S2FL [49]	EndNet [22]	CoupledCNN [50]	FGCN [18]	HCT [42]	Ours
C01	Trees	83.18±00.30	81.51±00.53	84.16±00.65	86.01±03.28	90.36±00.84	89.45±00.80	**91.77±00.72**
C02	Mostly Grass	84.20±00.28	76.86±02.24	83.23±01.63	87.26±01.29	**88.53±01.78**	86.02±02.38	85.51±01.10
C03	Mixed Ground Surface	68.72±00.05	69.24±02.01	70.86±02.14	77.66±06.62	82.04±00.54	81.44±01.78	**82.83±01.31**
C04	Dirt and Sand	84.85±00.08	78.59±02.98	85.70±01.89	93.60±03.97	93.73±01.45	94.88±01.72	**95.61±01.49**
C05	Road	88.00±00.39	84.21±01.66	87.77±01.10	89.59±02.11	**90.87±00.53**	84.10±01.57	86.71±00.90
C06	Water	93.59±00.16	93.53±00.76	94.34±01.96	98.08±00.32	100.00±00.00	99.64±00.38	**100.00±00.00**
C07	Buildings Shadow	86.23±00.26	84.16±01.68	87.74±01.07	90.83±02.81	85.76±01.32	94.54±01.32	**94.62±00.92**
C08	Buildings	83.39±00.11	76.33±01.23	81.73±01.58	95.40±01.63	92.08±01.47	**93.81±00.80**	93.23±01.64
C09	Sidewalk	75.62±00.46	71.53±03.57	75.37±02.96	69.94±03.37	78.40±07.44	77.84±02.64	**83.72±04.56**
C10	Yellow Curb	97.93±00.75	92.62±04.24	97.47±02.25	94.26±05.76	96.60±01.83	**98.21±01.15**	97.91±01.73
C11	Cloth Panels	97.98±00.73	84.91±02.43	97.64±00.96	98.32±00.18	100.00±00.00	99.78±00.21	**100.00±00.00**
OA (%)		81.57±00.11	77.62±00.42	82.84±00.49	87.22±01.34	89.04±00.38	88.47±00.81	**90.66±00.38**
AA (%)		85.63±00.15	81.15±03.03	85.94±00.38	89.14±02.28	90.76±00.21	90.89±00.22	**92.21±00.32**
Kappa (×100) (%)		77.63±00.09	72.69±00.62	78.72±00.75	84.52±01.77	86.13±00.73	85.19±00.79	**87.73±00.66**

TABLE IV
CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE HOUSTON2013 DATASET

No.	Class Name	SVM [12]	S2FL [49]	EndNet [22]	CoupledCNN [50]	FGCN [18]	HCT [42]	Ours
C01	Healthy Grass	95.44±00.57	97.82±00.84	96.13±00.80	93.64±03.77	98.72±00.07	99.48±00.51	**99.53±00.13**
C02	Stressed Grass	97.85±02.16	87.11±02.38	94.72±00.87	96.86±03.89	98.30±00.39	99.84±00.15	**99.91±00.09**
C03	Synthetic Grass	100.00±00.00	99.02±00.41	98.73±00.72	99.39±00.55	98.87±00.49	99.84±00.16	**100.00±00.00**
C04	Trees	94.61±02.01	92.20±03.34	91.50±03.39	97.37±02.75	98.54±00.51	99.74±00.24	**100.00±00.00**
C05	Soil	99.11±00.19	98.50±00.52	98.98±00.38	99.30±00.60	99.91±00.30	100.00±00.00	**100.00±00.00**
C06	Water	96.92±01.08	84.83±02.68	93.22±00.87	90.25±07.58	100.00±00.00	99.99±00.03	**100.00±00.00**
C07	Residential	92.11±08.19	83.18±02.82	89.25±01.12	95.89±02.60	98.15±00.62	**99.64±00.57**	99.47±00.46
C08	Commercial	93.81±04.45	53.25±03.33	76.52±02.06	92.49±04.94	97.68±00.90	99.43±00.79	**100.00±00.00**
C09	Road	88.34±01.12	80.52±01.24	86.05±01.09	90.40±03.89	97.27±00.38	**99.88±00.27**	99.53±00.15
C10	Highway	94.13±01.77	73.11±01.89	91.28±01.12	96.53±02.17	98.87±01.14	100.00±00.00	**100.00±00.00**
C11	Railway	95.79±09.10	64.75±02.85	87.60±01.83	98.28±01.27	97.58±01.70	99.98±00.04	**100.00±00.00**
C12	Paking Lot1	88.48±05.76	69.24±03.17	83.18±01.31	91.01±05.68	98.18±00.58	99.66±00.35	**99.90±00.09**
C13	Paking Lot2	81.66±04.64	43.51±01.96	84.18±02.27	92.04±06.69	98.97±04.88	**99.72±00.68**	99.65±00.82
C14	Tennis Court	99.53±00.13	98.95±00.47	98.33±00.60	99.74±00.38	100.00±00.00	100.00±00.00	**100.00±00.00**
C15	Running Track	99.70±00.17	99.18±00.41	98.99±00.38	99.89±00.25	100.00±00.00	100.00±00.00	**100.00±00.00**
OA (%)		93.34±05.30	81.26±00.39	90.18±00.87	95.48±00.88	98.50±00.29	99.78±00.11	**99.81±00.05**
AA (%)		94.50±04.52	81.78±00.37	90.28±00.25	95.51±01.11	98.74±00.41	99.81±00.09	**99.86±00.06**
Kappa (×100) (%)		93.88±05.72	79.66±00.43	88.72±00.34	95.11±00.95	98.38±00.32	99.77±00.12	**99.80±00.09**

layers from the set {0, 1, 2, 3, 4, 5} to evaluate their impact. Fig. 5(e) and (f) shows the OA of MACN on six datasets with different MACT and MCGF layers. The experimental results indicate that the optimal OA value was achieved on all six datasets when both MACT and MCGF layers were set to 1.

D. Classification Results and Analysis

To validate the effectiveness of MACN in classification tasks, we conducted comparative experiments with several representative classification methods, including SVM [12], shared and specific feature learning (S2FL) [48], EndNet [22], CoupledCNN [49], fractional gabor convolutional network (FGCN) [18], and hierarchical CNN and transformer (HCT) [42]. For fair comparison, the network parameters of these methods were set according to the descriptions in their respective reference papers.

Tables III–VI show the OA, AA, Kappa, and accuracy for each class of the proposed method on the MUUFL, Houston2013, Houston2018, and Trento datasets. The best results are shown in bold. It can be observed that the proposed MACN performs well and achieves higher accuracy in terms of the three evaluation metrics compared with other methods.

Compared with HCT, which is also a Transformer architecture, the MACN achieved +2.19% OA, +1.32% AA, and +2.54% Kappa on the MUUFL dataset, and our parameter amount is only 60% of it. Specifically, we achieved the best results in seven out of 11 classes and improved the accuracy of $C09$ Sidewalk by 4.88%. Fig. 6 shows the results of the visualization in the MUUFL dataset. It can be seen from the figure that the features extracted by MACN exhibit good continuity in regions formed by the same material, and the boundaries between regions formed by different materials are sharper. Compared with other methods, our results are closer to the ground truth obtained from manual labeling.

On the Houston2013 dataset, we achieved 100% accuracy in nine out of 15 classes and obtained the OA, AA, and Kappa of 99.81% ± 00.05, 99.86% ± 00.06, and 99.80% ± 00.09, respectively. For classes with similar materials, such as $C01$–$C04$, which are trees and different types of grass, the proposed method achieved the accuracies of 99.53%, 99.91%, 100%, and 100%. This demonstrates that the MACN can leverage the complementary advantages of multisource RS information to improve classification accuracy. Moreover, for

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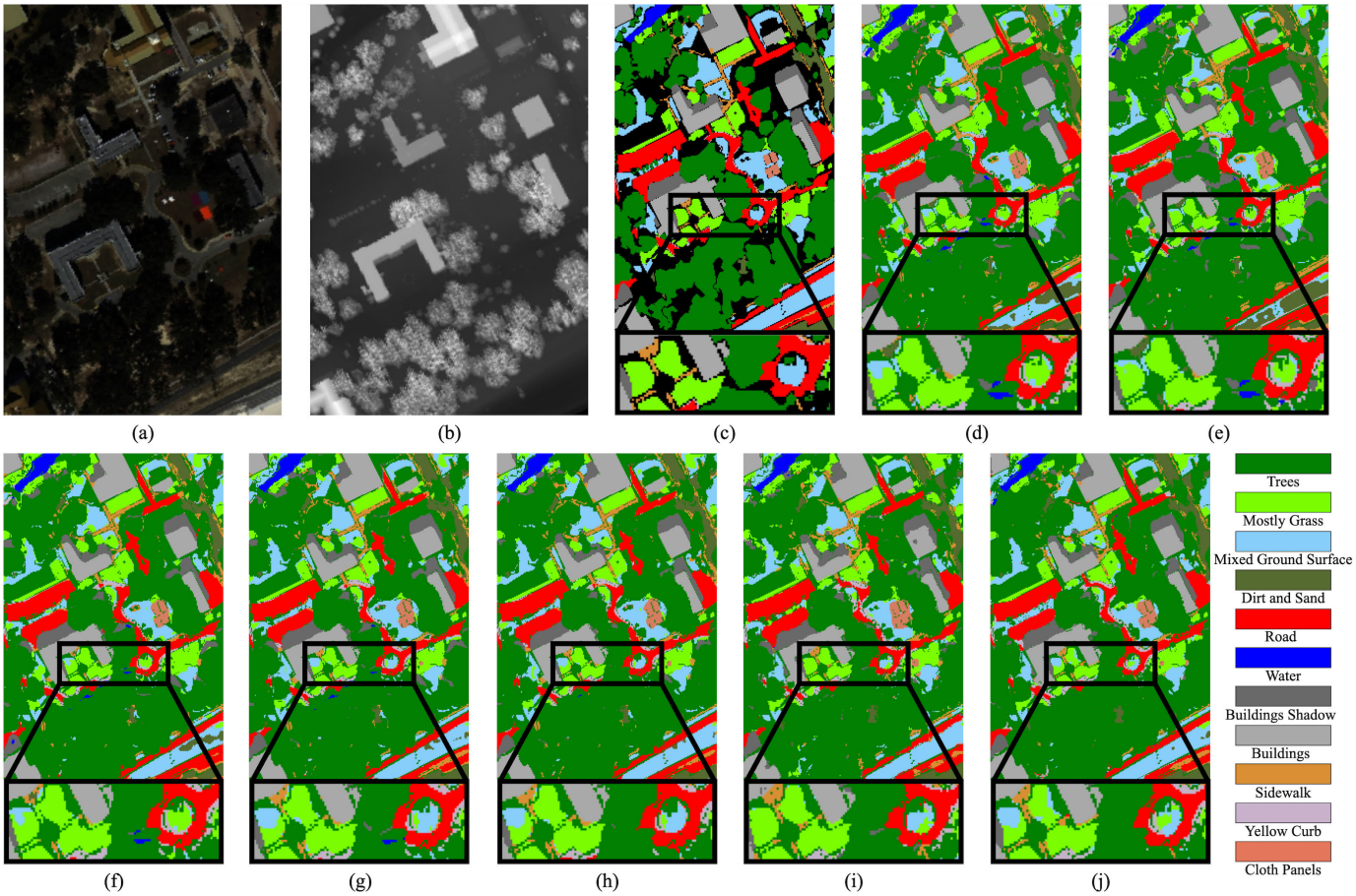


Fig. 6. Classification maps by different methods on the MUUFL dataset. OA—(a) pseudo-color image for HSI, (b) LiDAR-based DSM, (c) ground-truth map, (d) SVM (81.57%), (e) S2FL (77.62%), (f) EndNet (82.84%), (g) CoupledCNN (87.2%), (h) FGCN (89.04%), (i) HCT (88.47%), and (j) ours (90.66%).

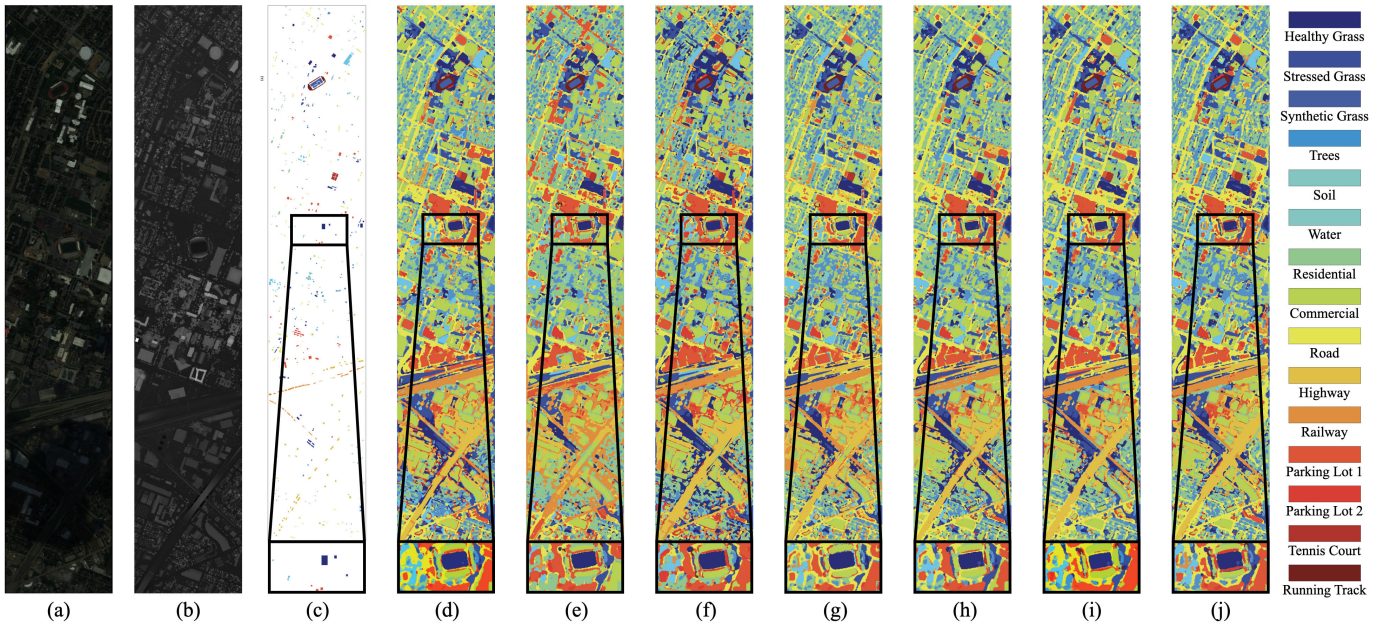


Fig. 7. Classification maps by different methods on the Houston2013 dataset. OA—(a) pseudo-color image for HSI, (b) LiDAR-based DSM, (c) ground-truth map, (d) SVM (93.34%), (e) S2FL (81.26%), (f) EndNet (90.18%), (g) CoupledCNN (95.48%), (h) FGCN (98.50%), (i) HCT (99.78%), and (j) ours (99.81%).

the central sports field region in Fig. 7, which has a large scale difference compared to other regions, the proposed method can obtain good classification results. This is because MACT can extract multiscale features that can maintain sensitivity to local semantic changes while obtaining global features.

On the Houston2018 dataset, we achieved the OA, AA, and Kappa of 84.58% ± 01.92, 87.65% ± 01.57, and 78.41% ± 02.43, respectively. Comparing with the Houston2013 dataset, we found that the Houston2018 dataset is more challenging. The visualization results of the Houston2018 dataset are

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TABLE V
CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE HOUSTON2018 DATASET

No.	Class Name	SVM [12]	S2FL [49]	EndNet [22]	CoupledCNN [50]	FGCN [18]	HCT [42]	Ours
C01	Healthy Grass	33.08±03.91	50.42±02.43	70.95+04.13	79.86±02.43	60.42±00.87	82.32±02.81	**84.75±01.45**
C02	Stressed Grass	74.71±00.12	57.72±01.95	82.53+02.41	87.71±03.63	77.71±03.45	90.00±02.28	**90.78±01.29**
C03	Artificial Turf	84.70±02.76	74.62±03.11	89.09+01.76	98.30±02.46	86.61±02.92	98.30±01.13	**100.00±00.00**
C04	Evergreen Trees	52.36±02.76	68.39±02.98	83.41+03.59	88.28±03.37	78.38±01.76	93.67±02.02	**93.85±02.15**
C05	Deciduous Trees	46.16±03.22	63.22±04.04	78.71+02.11	83.48±02.51	72.22±03.23	84.86±01.41	**86.30±01.87**
C06	Bare Earth	77.53±01.89	88.46±03.36	96.38+04.35	98.51±02.48	92.45±01.58	99.64±01.76	**99.71±00.63**
C07	Water	21.01±00.93	83.96±01.53	92.01+02.74	96.64±03.99	83.95±02.14	**99.16±00.94**	98.31±02.92
C08	Residential	63.41±02.39	79.92±04.05	82.39+03.98	87.00±02.93	81.92±03.01	93.40±01.09	**93.79±00.94**
C09	Non-Residential	53.55±03.11	21.53±02.94	44.89+02.88	48.84±02.10	34.53±02.67	**59.37±03.51**	58.91±02.79
C10	Roads	38.01±02.14	29.54±04.48	61.10+02.82	65.60±01.12	39.53±01.42	70.40±02.33	**73.48±01.36**
C11	Sidewalks	21.79±02.39	34.85±02.43	59.10+03.42	60.26±02.72	36.84±02.31	68.46±01.44	**70.33±03.29**
C12	Crosswalks	05.00±00.17	8.53±03.56	36.45+03.79	42.38±03.99	12.52±01.12	62.57±02.01	**68.91±02.68**
C13	Major Thoroughfares	53.94±03.54	69.82±04.03	86.22+02.25	90.52±01.95	69.82±02.79	**92.71±03.13**	89.92±02.77
C14	Highways	10.02±01.63	58.13±03.23	80.45+04.53	81.36±00.61	60.12±03.95	89.92±02.76	**90.46±01.54**
C15	Railways	76.07±01.75	69.35±02.52	84.08+03.14	94.89±02.15	74.34±02.03	96.13±01.54	**97.00±03.28**
C16	Paved Parking Lots	30.59±01.42	61.85±04.94	77.04+02.06	82.09±02.47	72.85±01.28	83.80±02.10	**84.09±01.65**
C17	Unpaved Parking Lots	10.53±03.03	88.74±02.68	97.36+03.73	100.00±00.00	90.73±03.68	100.00±00.00	**100.00±00.00**
C18	Cars	39.74±02.51	65.82±03.08	71.89+04.04	76.21±02.63	63.81±02.44	**89.24±01.87**	89.11±02.13
C19	Trains	51.42±03.31	43.35±04.02	63.82+02.23	67.27±01.91	53.35±01.07	84.89±02.13	**84.92±01.46**
C20	Stadium Seats	88.57±01.66	71.77±02.13	89.33+03.19	97.21±02.17	78.76±02.95	98.44±01.38	**98.46±00.97**
OA (%)		47.24±01.87	53.62±02.70	71.29±03.24	75.19±01.97	68.40±02.12	82.31±02.30	**84.58±01.92**
AA (%)		46.61±02.43	60.99±03.23	76.36±03.19	81.32±02.44	66.04±02.30	86.86±01.92	**87.65±01.57**
Kappa (×100) (%)		44.59±01.62	51.56±03.64	65.52±02.99	68.43±02.81	58.34±03.48	72.77±02.17	**78.41±02.43**

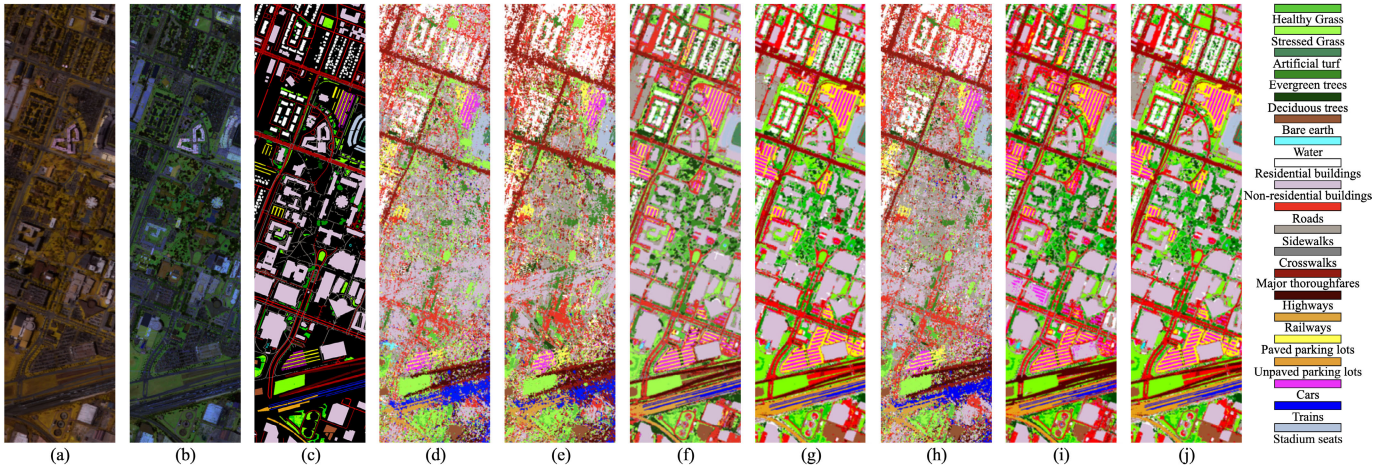


Fig. 8. Classification maps by different methods on the Houston2018 dataset. OA—(a) pseudo-color image for HSI, (b) pseudo-color image for MS-LiDAR, (c) ground-truth map, (d) SVM (47.24%), (e) S2FL (53.62%), (f) EndNet (71.29%), (g) CoupledCNN (75.19%), (h) FGCN (68.40%), (i) HCT (82.31%), and (j) ours (84.58%).

shown in Fig. 8. However, we still achieved 100% accuracy on the $C03$ Artificial Turf and $C17$ Unpaved Parking Lots classes and set the state-of-the-art (SOTA) performance for 16 classes. We observed limited classification accuracy for the classes $C10$ Roads, $C11$ Sidewalks, and $C12$ Crosswalks. This limitation might be attributed to the similarities in spectral and elevation information among these three classes, making them difficult to distinguish.

On the Trento dataset, the MACN achieved $99.73\% \pm 00.15$ OA, $99.67\% \pm 00.24$ AA, and $99.66\% \pm 00.17$ Kappa. The visualization results of the Trento dataset are shown in Fig. 9. It can be seen that we achieved a very good discrimination effect for areas with large differences in elevation information, such as $C04$ Woods and $C05$ Vineyard, achieving 100% accuracy. However, the results for $C03$ Ground were not the SOTA, and the reason for the poor classification effect is that the spectral reflectance curve of $C01$ Apple Trees adjacent to it is very similar, and the elevation is also similar, resulting in some slight misclassification between the two classes.

The above four datasets have validated the effectiveness of our proposed model on HSI and LiDAR data. In order to further explore the performance of the proposed model on other RS datasets, we conducted relevant experiments in Augsburg and Berlin.

From Table VII, it can be seen that FGCN performed poorly in the fusion of PolSAR and HSI, while our model once again demonstrated its powerful ability, achieving an increase of +45.97% in OA compared with FGCN. HCT also exhibited stable performance, but the proposed MACN generally improved the accuracy by 3.0%–4.43% in $C03$ Industrial Area, $C05$ Allotment, and $C07$ Water.

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TABLE VI
CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE TRENTO DATASET

No.	Class Name	SVM [12]	S2FL [49]	EndNet [22]	CoupledCNN [50]	FGCN [18]	HCT [42]	Ours
C01	Apple Trees	89.49±00.30	80.53±01.91	87.50±00.70	99.54±00.30	99.32±00.30	99.10±00.43	**99.64±00.32**
C02	Buildings	76.47±00.16	79.86±03.55	87.36±00.91	97.61±01.98	96.88±02.76	98.95±00.41	**99.57±00.56**
C03	Ground	97.70±00.91	90.21±01.71	98.23±00.91	98.34±00.19	96.04±05.03	**100.00±00.00**	99.96±00.27
C04	Woods	93.63±00.70	93.04±00.71	98.34±00.19	100.00±00.00	100.00±00.00	100.00±00.00	**100.00±00.00**
C05	Vineyard	73.53±00.36	82.58±01.26	93.68±00.26	100.00±00.00	99.68±00.07	99.99±00.01	**100.00±00.00**
C06	Roads	73.91±00.60	80.32±02.49	86.59±01.29	97.69±01.44	98.78±02.38	98.45±01.22	**98.84±01.09**
OA (%)		82.44±00.13	85.11±00.56	92.92±00.12	98.79±00.38	99.32±00.30	99.60±00.11	**99.73±00.15**
AA (%)		84.12±00.17	84.43±00.62	91.78±00.21	97.70±00.37	98.45±00.41	99.34±00.17	**99.67±00.24**
Kappa (×100) (%)		77.03±00.14	80.21±00.72	90.53±00.16	98.38±00.50	99.09±01.05	99.46±00.15	**99.66±00.17**

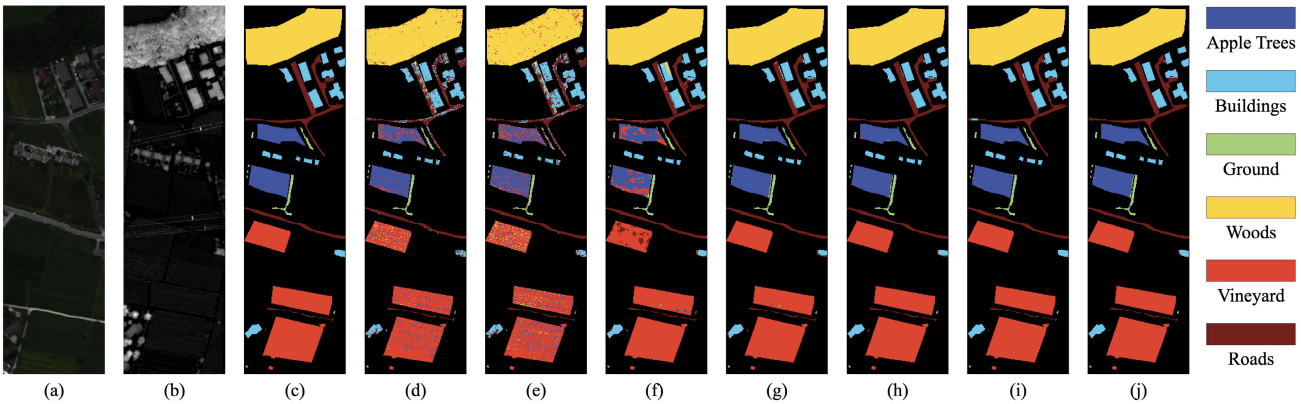


Fig. 9. Classification maps by different methods on the Trento dataset. OA—(a) pseudo-color image for HSI, (b) LiDAR-based DSM, (c) ground-truth map, (d) SVM (82.44%), (e) S2FL (85.11%), (f) EndNet (92.92%), (g) CoupledCNN (98.79%), (h) FGCN (99.32%), (i) HCT (99.60%), and (j) ours (99.73%).

TABLE VII
CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE AUGSBURG DATASET

No.	Class Name	SVM [12]	S2FL [49]	EndNet [22]	CoupledCNN [50]	FGCN [18]	HCT [42]	Ours
C01	Forest	94.35±00.89	89.68±00.32	94.32±00.85	98.22±00.92	78.85±00.55	97.93±00.55	**99.06±00.36**
C02	Residential Area	86.97±00.71	86.31±01.43	88.87±00.28	98.14±00.50	33.09±01.64	98.66±00.46	**99.13±00.46**
C03	Industrial Area	08.50±01.62	35.39±03.38	42.03±03.46	87.05±05.02	47.99±02.73	84.61±02.39	**89.04±03.18**
C04	Low Plants	64.75±01.40	88.38±00.96	94.58±00.10	98.36±00.88	57.77±00.38	98.80±00.73	**99.23±00.24**
C05	Allotment	06.58±02.55	66.63±05.43	32.33±02.03	80.26±05.14	81.60±05.27	85.21±05.44	**89.21±04.63**
C06	Commercial Area	08.82±01.10	09.96±02.48	28.98±04.78	65.00±04.09	56.26±04.51	**80.17±03.49**	73.70±03.91
C07	Water	48.64±02.34	47.38±01.78	53.01±01.63	68.64±06.72	70.08±03.62	63.14±02.73	**67.47±02.49**
OA (%)		72.89±00.73	81.63±00.44	86.99±00.31	95.18±00.30	51.59±00.30	96.81±00.23	**97.56±00.15**
AA (%)		45.52±00.53	62.00±01.02	62.01±00.62	85.10±02.45	60.81±00.41	86.64±00.90	**88.08±00.77**
Kappa (×100) (%)		62.19±00.85	74.54±00.56	81.21±00.38	94.95±00.59	38.89±01.05	95.58±00.78	**96.37±00.82**

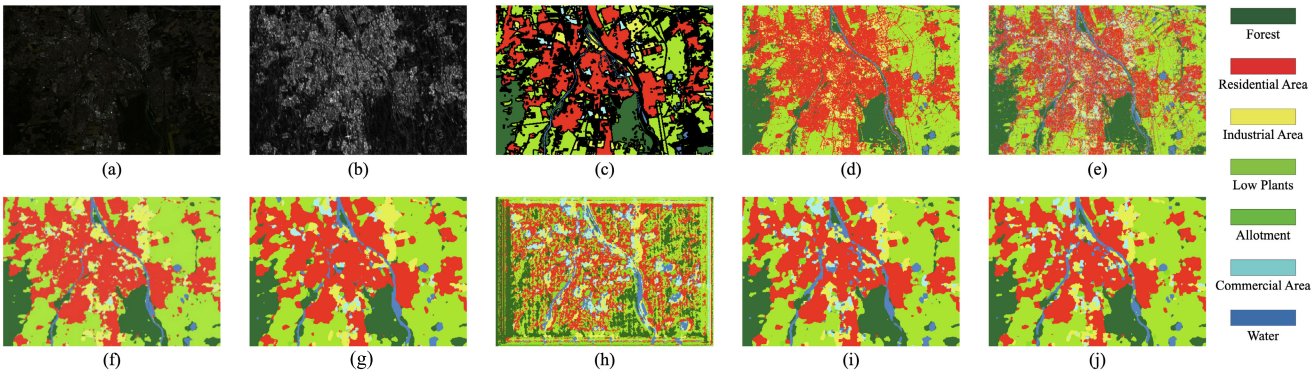


Fig. 10. Classification maps by different methods on the Augsburg dataset. OA—(a) pseudo-color image for HSI, (b) dual-Pol PolSAR, (c) ground-truth map, (d) SVM (72.89%), (e) S2FL (81.63%), (f) EndNet (86.99%), (g) CoupledCNN (95.18%), (h) FGCN (51.59%), (i) HCT (96.81%), and (j) ours (97.56%).

However, the performance of MCAN on $C06$ Commercial Area is mediocre. As shown in Fig. 10, $C06$ is located close to $C03$, and its PolSAR and HSI data have similar information, which leads to misclassification of Commercial Area as Industrial Area. It can be seen from Table VIII that MACN can also be well used to extract

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TABLE VIII
CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE BERLIN DATASET

No.	Class Name	SVM [12]	S2FL [49]	EndNet [22]	CoupledCNN [50]	FGCN [18]	HCT [42]	Ours
C01	Forest	88.45±00.52	83.30±00.91	89.50±00.25	**94.02±00.30**	04.73±03.13	92.21±00.38	90.24±00.28
C02	Residential Area	83.39±00.41	57.39±01.24	58.63±00.55	72.45±00.83	31.90±01.90	77.77±00.11	**81.15±00.48**
C03	Industrial Area	59.31±01.50	48.53±01.17	60.34±03.57	89.66±05.38	36.43±02.76	**91.36±02.35**	85.37±03.19
C04	Low Plants	86.08±01.03	77.16±01.51	88.28±00.66	90.90±00.40	57.13±00.64	91.50±00.21	**91.73±00.38**
C05	Soil	81.10±01.59	83.84±01.67	91.21±00.63	**97.14±00.59**	75.93±00.29	95.79±00.87	96.49±00.26
C06	Allotment	59.53±02.43	57.05±02.52	31.72±02.58	82.46±03.61	25.77±05.91	**85.79±05.24**	81.21±04.73
C07	Commercial Area	51.54±01.15	31.02±01.36	59.36±04.74	**64.53±03.11**	29.78±04.66	60.64±03.39	57.12±03.95
C08	Water	90.32±02.42	61.57±02.34	70.63±01.58	80.81±05.01	23.83±03.55	93.56±02.42	**94.13±02.13**
OA (%)		82.96±00.35	62.23±00.68	68.62±00.65	78.96±00.52	32.98±00.15	81.99±00.95	**84.30±00.26**
AA (%)		74.97±00.96	62.48±00.47	68.71±00.68	84.00±02.94	26.46±00.65	**86.08±00.41**	84.64±00.38
Kappa (×100) (%)		70.80±00.55	48.77±00.75	56.71±00.23	70.05±00.51	19.11±01.03	73.79±00.48	**74.87±00.53**

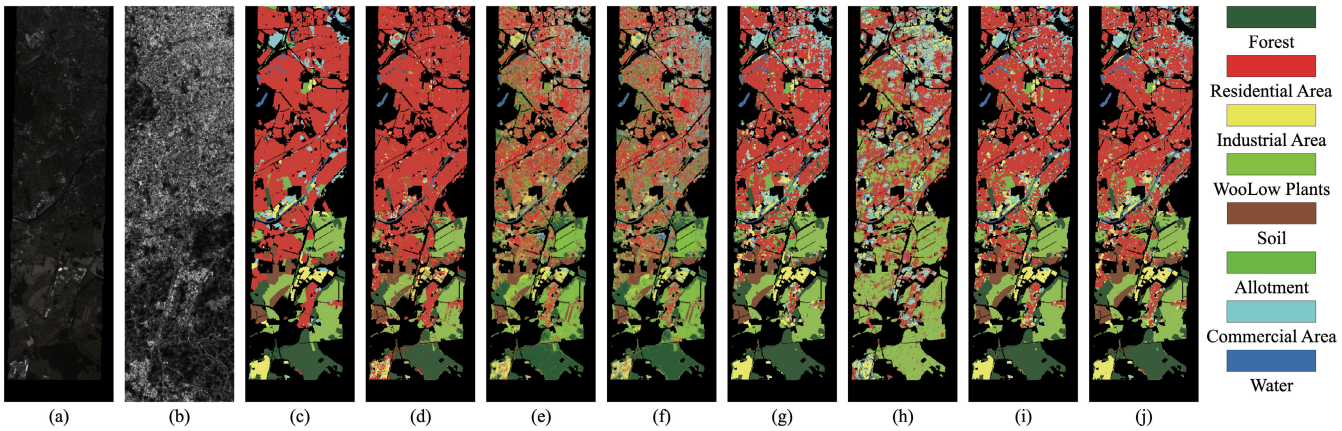


Fig. 11. Classification maps by different methods on the Berlin dataset. OA—(a) pseudo-color image for HSI, (b) SAR, (c) ground-truth map, (d) SVM (82.96%), (e) S2FL (62.23%), (f) EndNet (68.62%), (g) CoupledCNN (78.96%), (h) FGCN (32.98%), (i) HCT (81.99%), and (j) ours (84.30%).

SAR data features and achieve the SOTA OA in the Berlin dataset.

In general, the MACN can capture local–global multiscale information and maximize the complementary advantages of multisource RS data, resulting in a smoother classification result map.

As shown in Table IX, we compared the parameter size and computation time of our method with the CNN-based and Transformer-based model, including EndNet, CoupledCNN, FGCN, and HCT. Compared with the HCT model, which is also based on the Transformer architecture, our method has a smaller number of parameters and requires less time for training and testing. For example, on the MUUFL dataset, our method requires only 66% of the training time of HCT.

E. Ablation Analysis

1) To verify the impact of multisource data fusion on the classification performance of the model, we conducted six sets of experiments: single HSI data input, single LiDAR data input, single SAR data input, mixed HSI and LiDAR input, and mixed HSI and SAR input. The experimental results are shown in Table X. By comparing the OA, AA, and Kappa on the three datasets, the proposed MACN obtained better classification results. This confirms that the two types of RS data have a positive effect on improving classification performance and validates that MACN can fully utilize effective information from different modalities of RS data.
2) To verify the effectiveness of the components in the MACN, we conducted ablation experiments to analyze the contribution of each component to classification performance. Specifically, the proposed framework is mainly divided into four components: the ACE for extracting shallow CNN features, the MACT for extracting multiscale information, the MCGF for fusing multisource RS data features, and the loss function with dynamic weights.

We evaluated five variants of the proposed framework using the MUUFL dataset, as shown in Table XI. The classification accuracy of these variants was analyzed to evaluate the impact of the various components on the model. In Case1, we removed the ACE and used the basic Conv-2D (the upper branch in Fig. 2) to extract shallow CNN features. Compared with Case4, the OA decreased by 1.09%, which indicates that the proposed ACE can indeed better extract spectral–spatial information from RS data. This also validates that the success of MACN largely comes from MACT and MCGF. Case2 and Case3 removed MACT and MCGF, respectively, and decreased the OA by 2.21% and 2.19% compared with Case4. This suggests that using either MACT or MCGF alone can lead to a performance degradation, but using both together can achieve better classification accuracy. MACN added the dynamic

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TABLE IX
TRAINING TIME IN MINUTES (M) AND TEST TIME IN SECONDS (S) BETWEEN THE CONTRAST METHODS AND PROPOSED METHOD

Methods	#Param.	MUUFL		Houston2013		Houston2018		Trento		Augsburg		Berlin	
		Train.	Test	Train.	Test	Train.	Test	Train.	Test	Train.	Test	Train.	Test
EndNet	0.35 M	**1.73**	**4.18**	**1.54**	**2.98**	**5.82**	**10.56**	**2.58**	**4.91**	**1.48**	**12.47**	**1.43**	**40.25**
CoupledCNN	0.11 M	2.61	7.86	3.04	3.84	11.28	17.50	2.88	5.86	6.46	16.48	1.67	41.22
FGCN	1.43 M	4.01	7.92	5.64	4.37	22.90	30.86	4.39	6.28	12.77	26.34	3.04	67.30
HCT	0.43 M	7.73	8.16	7.25	4.91	27.23	33.13	5.32	6.43	15.88	39.79	3.96	73.29
Ours	0.33 M	5.12	8.03	6.03	4.63	25.14	31.85	4.88	6.34	13.26	28.91	3.55	69.32

TABLE X
ABLATION ANALYSIS OF DIFFERENT MODAL DATA INPUTS

Cases	Indicators	MUUFL	Houston2013	Houston2018	Trento	Augsburg	Berlin
Only HSI	OA (%)	88.37	99.27	75.83	99.38	96.04	82.62
	AA (%)	90.03	99.28	81.52	99.25	86.97	81.98
	Kappa (×100)(%)	86.50	99.16	67.91	99.19	95.40	72.35
Only LiDAR/SAR	OA (%)	50.28	65.71	71.23	92.02	67.32	57.83
	AA (%)	48.22	65.30	62.71	87.64	56.53	53.44
	Kappa (×100)(%)	49.04	64.87	59.35	89.25	66.78	47.03
HSI+LiDAR/SAR	OA (%)	**90.66**	**99.81**	**84.58**	**99.73**	**97.56**	**84.30**
	AA (%)	**92.21**	**99.86**	**87.65**	**99.67**	**88.08**	**84.64**
	Kappa (×100) (%)	**87.73**	**99.80**	**78.41**	**99.66**	**96.37**	**74.87**

TABLE XI
ABLATION ANALYSIS OF DIFFERENT COMPONENTS IN MACN ON THE MUUFL DATASET

Cases	MUUFL				Indicators		
	ACE	MACT	MCGF	Dyn. Loss	OA (%)	AA (%)	K×100(%)
1	–	✓	✓	–	89.08	89.94	85.57
2	✓	–	✓	–	87.96	90.31	85.63
3	✓	✓	–	–	87.38	89.22	84.74
4	✓	✓	✓	–	90.17	91.85	87.28
MACN	✓	✓	✓	✓	**90.66**	**92.21**	**87.73**

weight loss function on Case4, which improved the OA by 0.59%. This indicates that the imbalance of sample data in RS classification tasks can affect the accuracy of classification, and the proposed loss function can effectively alleviate the problem of sample imbalance. The ablation analysis further confirms the validity of the proposed network framework, showing that all of its major components are important.

IV. CONCLUSION

In this article, we proposed a novel unified framework, MACN, to solve RS classification tasks. By reanalyzing self-attention and convolution, we developed a new method for extracting local–global features called MACT, which is used to fully explore effective features. We designed an effective feature fusion module, MCGF, to achieve cross guidance between different modalities and improve fusion performance. Experiments are conducted on six multisource RS datasets to compare the method with the existing classification methods, and the results verify that the proposed method achieves effective and superior performance. In the future, we will try more types of RS data modalities to further verify the classification performance of our model. We will also explore whether MACN has good performance on tasks other than classification, such as segmentation, detection, and so on. In addition, we have found that there is an urgent need to propose new datasets for RS classification tasks. Therefore, we hope to contribute to the development of the RS community by proposing new datasets.

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